

**INTERNSHIP PROJECT REPORT**

# **Sales Performance Analysis and Interactive Business Intelligence Dashboard**

Data Analytics Internship

**Submitted by**  
**Manish Singh**

Tools: Microsoft Excel | SQL | Python | NumPy | Power BI

## **Internship Submission Report**

### **Internship Details**

**Applicant Name:** Manish Singh

**Application ID:** KTS020260812721

**Internship Domain:** Data Analytics Internship

**Duration:** 30 Days

**From:** 20 August 2026

**To:** 19 September 2026

## 1. Project Overview

This internship project focuses on analyzing a real-world sales dataset and converting raw transactional data into meaningful business intelligence. The project follows an end-to-end analytics workflow covering data cleaning, database storage, statistical analysis, visualization, and business interpretation.

The cleaned data was prepared using Microsoft Excel and stored in a SQL database for structured and persistent data management. Python and NumPy were used for numerical and descriptive analysis, while Power BI was used to develop an interactive dashboard for monitoring sales, profit, quantity, orders, customers, regional performance, product/category performance, and payment-method contribution.

## 2. Project Objectives

- Clean and preprocess the raw sales dataset.
- Store the cleaned dataset in a structured SQL database And Excel.
- Perform descriptive and numerical analysis using Python and NumPy.
- Identify yearly and monthly sales and profit trends.
- Evaluate category and product-level performance.
- Analyze regional sales and order performance.
- Understand sales-channel and customer purchasing behavior.
- Develop an interactive Power BI dashboard with relevant KPIs and visualizations.
- Generate actionable business insights and recommendations.

## 3. Dataset Description

| Parameter       | Value                                                             |
|-----------------|-------------------------------------------------------------------|
| Total Records   | 50,000                                                            |
| Total Sales     | ₹2,181,259,557                                                    |
| Total Quantity  | 149,676 units                                                     |
| Total Profit    | ₹194,163,285                                                      |
| Analysis Period | 2023–2026                                                         |
| Regions         | Central, East, North, South, West                                 |
| Main Categories | Electronics, Furniture, Fashion, Home Appliances, Office Supplies |
| Product Types   | 15                                                                |

The dataset contains transactional sales information with measures and dimensions such as Order\_ID, date, region, category, product, quantity, discount, unit price, sales, profit, profit margin, payment method, sales channel, and customer-related information.

## 4. Tools & Technologies

| Tool / Technology | Purpose                                                                     |
|-------------------|-----------------------------------------------------------------------------|
| Microsoft Excel   | Data cleaning, preprocessing, Pivot Tables and initial exploratory analysis |
| SQL               | Structured storage and retrieval of cleaned data                            |
| Python            | Data analysis and numerical processing                                      |
| NumPy             | Statistical and numerical analysis                                          |
| Power BI          | Interactive dashboard, KPI analysis and visualization                       |
| Jupyter Notebook  | Python analysis and documentation                                           |
| GitHub            | Source code, notebook and project documentation                             |

## 5. Data Cleaning & Preprocessing

The raw sales dataset was first inspected for data quality and consistency. Microsoft Excel was used to prepare the data for analysis. The cleaning stage focused on validating the dataset structure, checking duplicates and missing values, standardizing formats, and ensuring that numerical and categorical fields were appropriate for analysis.

### 5.1 Cleaning Workflow

- Inspect the raw dataset and identify data-quality issues.
- Check duplicate and missing records.
- Validate date, numerical and categorical fields.
- Standardize inconsistent values and formats where required.
- Remove or correct invalid records where applicable.
- Validate the cleaned dataset before analytical processing.

## 6. SQL Database Integration

After cleaning, the prepared dataset was stored in a SQL database. This provides a structured and persistent storage layer and makes the cleaned data available for future querying, analysis, reporting, and integration with analytics tools.

Workflow: Raw Data → Excel Cleaning → Clean Dataset → SQL Database → Python / Power BI → Insights

SQL was used as the database layer for maintaining the cleaned dataset. The database can also support future extensions such as automated data pipelines, scheduled refreshes, and SQL-based analytical queries.

## 7. Exploratory Data Analysis

Pivot-based analysis was performed to understand sales, profit, orders, categories, products, regions, and channels before developing the final dashboard.

### 7.1 Year-wise Sales Analysis

| Year | Sales    |
|------|----------|
| 2023 | ₹594.73M |
| 2024 | ₹607.94M |
| 2025 | ₹608.13M |
| 2026 | ₹370.46M |

Sales increased from 2023 to 2025, with 2025 recording approximately ₹608.13M. The 2026 value should be interpreted according to the available reporting period; if it represents a partial year, it should not be compared directly with complete years.

### 7.2 Year-wise Profit Analysis

| Year | Profit  |
|------|---------|
| 2023 | ₹53.10M |
| 2024 | ₹53.98M |
| 2025 | ₹54.00M |
| 2026 | ₹33.09M |

### 7.3 Monthly Sales Analysis

| Month | Sales    |
|-------|----------|
| Jan   | ₹203.91M |
| Feb   | ₹190.11M |
| Mar   | ₹196.49M |
| Apr   | ₹210.30M |
| May   | ₹198.79M |
| Jun   | ₹193.09M |
| Jul   | ₹194.19M |
| Aug   | ₹185.30M |
| Sep   | ₹141.69M |
| Oct   | ₹152.82M |
| Nov   | ₹160.81M |
| Dec   | ₹153.76M |

April recorded the highest monthly sales at approximately ₹210.30M, while September recorded the lowest at approximately ₹141.69M. The pattern indicates stronger sales in the first and middle portions of the year and a noticeable decline from September onward.

### 7.4 Monthly Profit Analysis

| Month | Profit  |
|-------|---------|
| Jan   | ₹18.35M |
| Feb   | ₹17.12M |
| Mar   | ₹17.53M |
| Apr   | ₹18.59M |
| May   | ₹17.73M |
| Jun   | ₹17.52M |
| Jul   | ₹17.39M |
| Aug   | ₹16.20M |
| Sep   | ₹12.45M |
| Oct   | ₹13.56M |
| Nov   | ₹13.87M |
| Dec   | ₹13.85M |

April generated the highest monthly profit at approximately ₹18.59M, while September recorded the lowest at approximately ₹12.45M. Profit broadly follows the sales pattern.

## 8. Category Performance Analysis

| Category        | Sales      | Profit   | Orders |
|-----------------|------------|----------|--------|
| Electronics     | ₹1,659.46M | ₹128.33M | 17,992 |
| Furniture       | ₹212.08M   | ₹26.28M  | 9,976  |
| Home Appliances | ₹175.98M   | ₹18.42M  | 8,052  |
| Office Supplies | ₹101.72M   | ₹16.54M  | 7,866  |
| Fashion         | ₹32.02M    | ₹4.59M   | 6,114  |

Electronics is the dominant category, contributing approximately ₹1.66B in sales and ₹128.33M in profit. It also records the highest order volume with 17,992 orders. This makes Electronics a critical category for inventory, pricing, promotion, and profitability decisions.

## 9. Product Performance Analysis

| Product  | Sales    |
|----------|----------|
| Laptops  | ₹962.07M |
| Mobiles  | ₹657.26M |
| Tables   | ₹98.48M  |
| Printing | ₹96.15M  |
| Storage  | ₹67.57M  |

**Laptops** are the highest-selling product at approximately ₹962.07M, followed by Mobiles at approximately ₹657.26M. These products represent major revenue contributors and should receive appropriate attention in inventory and sales planning.

## 10. Regional Sales Analysis

| Region       | Sales           | Orders        |
|--------------|-----------------|---------------|
| Central      | ₹275.26M        | 6,356         |
| East         | ₹367.18M        | 8,440         |
| <b>North</b> | <b>₹632.19M</b> | <b>14,399</b> |
| South        | ₹459.68M        | 10,532        |
| West         | ₹446.96M        | 10,273        |

**North** is the strongest region, generating approximately ₹632.19M in sales and 14,399 orders. South and West also show substantial contribution, while Central has the lowest sales and order volume among the five regions.

## 11. Sales Channel & Customer Purchasing Analysis

The dataset includes Marketplace, Online, and Retail Store sales channels. Channel-level order analysis was used to understand purchasing activity and identify differences in customer behavior across sales channels.

The Power BI dashboard further supports analysis through interactive filtering and payment-method visualization, allowing business users to explore how customers transact and how payment methods contribute to profit.

## 12. Python & NumPy Analysis

Python was used to perform numerical analysis and validate key business metrics. NumPy-based analysis and descriptive statistics provided a quantitative view of the dataset.

| Metric                    | Result            |
|---------------------------|-------------------|
| Total Sales               | ₹2,181,259,557.65 |
| Total Profit              | ₹191,549,820.35   |
| Total Quantity            | 149,676           |
| Average Quantity          | 2.99              |
| Average Discount          | 0.102             |
| Average Unit Price        | ₹16,139.24        |
| Average Sales per Record  | ₹43,625.19        |
| Average Profit per Record | ₹3,830.996        |
| Average Profit Margin     | 0.10877           |

## 12.1 Descriptive Statistics

| Variable      | Min         | Median     | Max         |
|---------------|-------------|------------|-------------|
| Quantity      | 1           | 2          | 10          |
| Discount      | 0           | 0.10       | 0.30        |
| Unit Price    | ₹162.04     | ₹6,955.83  | ₹71,497.92  |
| Sales         | ₹120.71     | ₹13,728.88 | ₹711,167.90 |
| Profit        | -₹17,705.84 | ₹1,265.34  | ₹104,456.85 |
| Profit Margin | -0.0419     | 0.1087     | 0.2638      |

The minimum profit is negative, indicating that the dataset contains loss-making transactions. The maximum profit is approximately ₹104,456.85, demonstrating substantial variation in transaction-level profitability.

## 13. Power BI Dashboard Development

An interactive Power BI dashboard was developed to convert the analytical results into a business-friendly visual reporting interface. The dashboard combines KPI cards, trend charts, regional analysis, category analysis, and payment-method analysis.

### 13.1 Dashboard KPIs

| KPI             | Value                    |
|-----------------|--------------------------|
| Total Sales     | ₹2,181.26M               |
| Total Profit    | ₹194.55M                 |
| Total Quantity  | 149.68K                  |
| Profit Margin   | 10.88%                   |
| Total Orders    | Verified dashboard value |
| Total Customers | Verified dashboard value |

The KPI values should reflect the final corrected Power BI model after the reconciliation of aggregation logic and data refresh. Total profit and order count were verified/fixed before final submission.

### 13.2 Dashboard Visualizations

- Monthly Sales – identifies monthly revenue trends.
- Monthly Profit – tracks monthly profitability.
- Sales Region Wise – compares sales-channel performance across regions.
- Profit Region Wise – compares profitability across regions.
- Category Wise Sales – identifies major revenue-generating categories.
- Profit by Payment Method – compares profit contribution across payment methods.

- KPI Cards – provide a quick summary of core business metrics.

## 14. Key Business Insights

- The dataset contains 50,000 transactional records and generates approximately ₹2.18B in total sales.
- Electronics is the strongest category, contributing approximately ₹1.66B in sales and ₹128.33M in profit.
- Laptops are the highest-selling product at approximately ₹962.07M, followed by Mobiles at approximately ₹657.26M.
- North is the highest-performing region with approximately ₹632.19M in sales and 14,399 orders.
- April recorded the highest monthly sales at approximately ₹210.30M and the highest monthly profit at approximately ₹18.59M.
- September recorded the lowest monthly sales at approximately ₹141.69M and the lowest monthly profit at approximately ₹12.45M.
- The presence of negative-profit transactions indicates that some sales are loss-making and may require further investigation.
- Customer purchasing activity can be evaluated through sales channels and payment methods available in the dashboard.

## 15. Business Recommendations

- Prioritize inventory availability and sales planning for Electronics, Laptops, and Mobiles because of their strong revenue contribution.
- Investigate loss-making transactions to identify whether discounts, pricing, procurement cost, or other factors are reducing profitability.
- Analyze the September sales decline and consider targeted promotions or demand-generation strategies during weaker months.
- Use the North region as a benchmark for identifying practices that could be applied to lower-performing regions.
- Develop category-specific marketing strategies for lower-performing categories such as Fashion.
- Use sales-channel and payment-method insights to optimize channel-specific customer engagement and transaction strategies.
- Use monthly sales and profit trends to improve inventory planning, staffing, and promotional scheduling.

## 16. Challenges & Solutions

| Challenge                    | Approach / Solution                                                         |
|------------------------------|-----------------------------------------------------------------------------|
| Large transactional dataset  | Used Pivot Tables and programmatic analysis for aggregation.                |
| Data quality and consistency | Performed cleaning and validation before analysis.                          |
| Multiple business dimensions | Analyzed time, category, product, region and channel dimensions separately. |
| Need for structured storage  | Stored the cleaned dataset in SQL.                                          |
| Complex business information | Converted analytical results into interactive Power BI visuals.             |
| Finding actionable patterns  | Compared sales, profit, orders and product/region performance.              |

## 17. End-to-End Project Workflow

Raw Sales Data → Excel Data Cleaning → Cleaned Dataset → SQL Database → Python/NumPy Analysis → Power BI Data Model & Dashboard → Business Insights → Recommendations

## 18. Learning Outcomes

- Practical experience in data cleaning and preprocessing.
- Hands-on use of Excel Pivot Tables for business analysis.
- Understanding of SQL-based data storage and retrieval.
- Practical Python and NumPy experience for numerical analysis.
- Experience in developing KPI-driven Power BI dashboards.
- Improved understanding of data visualization and business intelligence.
- Ability to convert analytical results into business recommendations.

## 19. Conclusion

The Sales Performance Analysis project demonstrates an end-to-end data analytics workflow from raw data preparation to business intelligence reporting. The dataset was cleaned using Excel, stored in SQL, analyzed using Python and NumPy, and transformed into an interactive Power BI dashboard. The analysis identified key sales and profit trends, leading product and category contributors, regional performance differences, and customer purchasing dimensions. The resulting insights can support inventory planning, product prioritization, regional strategy, promotional planning, and data-driven decision making.

## 20. Future Scope

- Integrate the SQL database directly with Power BI for a more automated reporting pipeline.
- Implement scheduled data refresh and automated dashboard updates.
- Develop sales and profit forecasting models.
- Perform customer segmentation and customer lifetime value analysis.
- Build predictive models for sales and profitability.
- Introduce automated business alerts for unusual sales or profit movements.

## 21. GitHub Repository & Deliverables

The final project package can include the public GitHub repository, Jupyter Notebook/source code, Power BI dashboard file, README documentation, and this Business Insights Report. The repository URL should be inserted below before submission:

GitHub Repository: \_\_\_\_\_

Dashboard File: SALES\_DATA\_POWER\_BI

Numpy Notebook(File Name): CDA

## 22. Appendix

### Appendix A — Excel Data Cleaning

| Order ID  | Order Date | Order Item        | Customer ID       | Customer Name   | Category    | Sub-Category           | Product Name | Region         | State       | City | Quantity | Discount | Unit Price | Sales    | Profit      | Payment Method | Customer Segment | Sales Channel | Year | Quarter | Month |
|-----------|------------|-------------------|-------------------|-----------------|-------------|------------------------|--------------|----------------|-------------|------|----------|----------|------------|----------|-------------|----------------|------------------|---------------|------|---------|-------|
| ORD105811 | 01-01-2023 | 001658 CUST11884  | Arijun Bansal     | Electronics     | Accessories | Power Bank 20000mAh    | East         | Bihar          | Patna       | 1    | 0.1      | 1680.68  | 1512.61    | 109.21   | Credit Card | Consumer       | Online           | 2023          | Q1   | Jan     |       |
| ORD130424 | 01-01-2023 | 002328 CUST17134  | Sneha Agarwal     | Electronics     | Accessories | Power Bank 20000mAh    | South        | Telangana      | Hyderabad   | 4    | 0.2      | 1669.86  | 5343.55    | 109.04   | UPI         | Corporate      | Retail Store     | 2023          | Q1   | Jan     |       |
| ORD144482 | 01-01-2023 | 004650 CUST13701  | Nandini Verma     | Furniture       | Storage     | Bookshelf              | North        | Uttarakhand    | Dehradun    | 8    | 0.05     | 5468.64  | 41561.66   | 4929.42  | Net Banking | Small Business | Online           | 2023          | Q1   | Jan     |       |
| ORD142947 | 01-01-2023 | 005217 CUST15891  | Manish Pandey     | Furniture       | Storage     | Filing Cabinet         | East         | Odisha         | Bhubaneswar | 2    | 0        | 6929.25  | 13858.5    | 2138.52  | Debit Card  | Corporate      | Online           | 2023          | Q1   | Jan     |       |
| ORD128842 | 01-01-2023 | 011551 CUST14335  | Neha Malhotra     | Electronics     | Mobiles     | iPhone 15              | South        | Karnataka      | Bengaluru   | 1    | 0.05     | 69925.84 | 66429.55   | 7446.75  | UPI         | Consumer       | Retail Store     | 2023          | Q1   | Jan     |       |
| ORD123361 | 01-01-2023 | 013211 CUST15049  | Sneha Das         | Electronics     | Laptops     | HP Pavilion 15         | West         | Goa            | Panaji      | 2    | 0        | 62740.95 | 125481.9   | 15095.47 | Net Banking | Corporate      | Online           | 2023          | Q1   | Jan     |       |
| ORD130591 | 01-01-2023 | 030014 CUST16003  | Raj Verma         | Electronics     | Mobiles     | OnePlus Nord CE        | West         | Goa            | Panaji      | 4    | 0.15     | 29902.69 | 101689.2   | 4036.10  | Debit Card  | Consumer       | Online           | 2023          | Q1   | Jan     |       |
| ORD149004 | 01-01-2023 | 035237 CUST17928  | Rahul Kumar       | Fashion         | Footwear    | Running Shoes          | North        | Punjab         | Ludhiana    | 3    | 0        | 3458.72  | 10376.16   | 1656.01  | Net Banking | Corporate      | Retail Store     | 2023          | Q1   | Jan     |       |
| ORD116197 | 01-01-2023 | 035418 CUST10899  | Rahul Mehta       | Furniture       | Tables      | Study Table            | East         | Jharkhand      | Ranchi      | 1    | 0.15     | 6397.23  | 5437.65    | 400.93   | EMI         | Consumer       | Marketplace      | 2023          | Q1   | Jan     |       |
| ORD126495 | 01-01-2023 | 044932 CUST16374  | Simran Agarwal    | Furniture       | Chairs      | Ergonomic Office Chair | West         | Gujarat        | Ahmedabad   | 2    | 0.15     | 8897.18  | 15125.21   | 1427.76  | Credit Card | Consumer       | Online           | 2023          | Q1   | Jan     |       |
| ORD134557 | 01-01-2023 | 052007 CUST13140  | Vivekan Bansal    | Office Supplies | Stationery  | Premium Notebook       | East         | Bihar          | Patna       | 1    | 0.15     | 484.83   | 412.11     | 60.53    | Cash        | Consumer       | Online           | 2023          | Q1   | Jan     |       |
| ORD139317 | 01-01-2023 | 054027 CUST15915  | Raj Choudhary     | Home Appliance  | Kitchen     | Air Fryer              | West         | Goa            | Panaji      | 4    | 0.2      | 6762.55  | 21640.16   | 951.99   | Debit Card  | Consumer       | Retail Store     | 2023          | Q1   | Jan     |       |
| ORD114546 | 01-01-2023 | 065030 CUST17226  | Simran Sharma     | Fashion         | Footwear    | Running Shoes          | East         | West Bengal    | Kolkata     | 3    | 0        | 3300.5   | 9901.5     | 2354.88  | Debit Card  | Corporate      | Online           | 2023          | Q1   | Jan     |       |
| ORD144551 | 01-01-2023 | 072052 CUST12944  | Neha Mishra       | Home Appliance  | Kitchen     | Air Fryer              | West         | Maharashtra    | Mumbai      | 10   | 0.05     | 7817.35  | 74264.82   | 7816.13  | Debit Card  | Small Business | Online           | 2023          | Q1   | Jan     |       |
| ORD143434 | 01-01-2023 | 085837 CUST15409  | Ananya Joshi      | Furniture       | Tables      | Office Work Desk       | South        | Telangana      | Hyderabad   | 1    | 0        | 12395.31 | 12395.31   | 2513.53  | Credit Card | Consumer       | Marketplace      | 2023          | Q1   | Jan     |       |
| ORD124232 | 01-01-2023 | 090514 CUST15975  | Pooja Bansal      | Electronics     | Laptops     | HP Pavilion 15         | East         | Bihar          | Patna       | 1    | 0.05     | 56756.51 | 53918.68   | 6491.81  | Debit Card  | Consumer       | Online           | 2023          | Q1   | Jan     |       |
| ORD105295 | 01-01-2023 | 095413 CUST11506  | Rohan Joshi       | Fashion         | Women       | Women's Kurti          | West         | Maharashtra    | Mumbai      | 2    | 0        | 1242.97  | 2485.94    | 450.48   | Cash        | Consumer       | Marketplace      | 2023          | Q1   | Jan     |       |
| ORD143144 | 01-01-2023 | 100015 CUST14352  | Prity Mehta       | Home Appliance  | Kitchen     | Air Fryer              | South        | Andhra Pradesh | Vijayawada  | 1    | 0.15     | 7031.7   | 9974.94    | 373.65   | Debit Card  | Consumer       | Online           | 2023          | Q1   | Jan     |       |
| ORD138101 | 01-01-2023 | 102218 CUST15812  | Riya Sharma       | Electronics     | Accessories | Bluetooth Speaker      | South        | Kerala         | Kochi       | 1    | 0.2      | 3741.49  | 2993.19    | 50.58    | UPI         | Corporate      | Online           | 2023          | Q1   | Jan     |       |
| ORD135800 | 01-01-2023 | 102712 CUST14895  | Vivekan Sharma    | Office Supplies | Stationery  | Ball Pen Pack          | West         | Gujarat        | Surat       | 2    | 0.1      | 189.86   | 341.75     | 60.83    | UPI         | Consumer       | Retail Store     | 2023          | Q1   | Jan     |       |
| ORD137046 | 01-01-2023 | 104529 CUST14483  | Manish Malhotra   | Electronics     | Accessories | Wireless Earbuds       | North        | Uttar Pradesh  | Kanpur      | 4    | 0.3      | 2188.6   | 6128.08    | 207.74   | Credit Card | Consumer       | Online           | 2023          | Q1   | Jan     |       |
| ORD135681 | 01-01-2023 | 133314 CUST14149  | Prity Malhotra    | Electronics     | Laptops     | Lenovo IdeaPad Slim    | West         | Gujarat        | Surat       | 3    | 0.05     | 5612.42  | 159963.2   | 13484.89 | Credit Card | Consumer       | Retail Store     | 2023          | Q1   | Jan     |       |
| ORD103041 | 01-01-2023 | 133334 CUST17122  | Neha Kumar        | Office Supplies | Paper       | A4 Paper Ream          | North        | Uttar Pradesh  | Kanpur      | 2    | 0.05     | 378.71   | 719.55     | 150.09   | UPI         | Consumer       | Online           | 2023          | Q1   | Jan     |       |
| ORD135905 | 01-01-2023 | 153454 CUST13836  | Arijun Choudhary  | Electronics     | Accessories | Bluetooth Speaker      | South        | Tamil Nadu     | Chennai     | 5    | 0        | 3540.61  | 17703.05   | 1975.66  | Credit Card | Consumer       | Retail Store     | 2023          | Q1   | Jan     |       |
| ORD104535 | 01-01-2023 | 154623 CUST13644  | Prity Joshi       | Home Appliance  | Kitchen     | Air Fryer              | South        | Andhra Pradesh | Vijayawada  | 4    | 0.1      | 6795.2   | 24354.72   | 2534.15  | Net Banking | Consumer       | Marketplace      | 2023          | Q1   | Jan     |       |
| ORD107684 | 01-01-2023 | 170642 CUST13207  | Aisha Jain        | Office Supplies | Printing    | Laser Printer          | North        | Uttarakhand    | Dehradun    | 4    | 0.15     | 19981.95 | 67938.63   | 10866.94 | Debit Card  | Small Business | Marketplace      | 2023          | Q1   | Jan     |       |
| ORD101194 | 01-01-2023 | 170742 CUST11188  | Pooja Bansal      | Furniture       | Chairs      | Ergonomic Office Chair | South        | Tamil Nadu     | Chennai     | 6    | 0.25     | 7840.44  | 35281.98   | 1349.09  | Cash        | Consumer       | Online           | 2023          | Q1   | Jan     |       |
| ORD101232 | 01-01-2023 | 183056 CUST16111  | Ankit Kumar       | Electronics     | Laptops     | Dell Inspiron 15       | South        | Telangana      | Hyderabad   | 1    | 0.15     | 61145.54 | 51973.71   | 2395.99  | Debit Card  | Consumer       | Retail Store     | 2023          | Q1   | Jan     |       |
| ORD137757 | 01-01-2023 | 200543 CUST16767  | Pooja Srivastava  | Fashion         | Men         | Men's Casual Shirt     | South        | Kerala         | Kochi       | 4    | 0.25     | 1355.6   | 4900.8     | 242.21   | EMI         | Corporate      | Online           | 2023          | Q1   | Jan     |       |
| ORD127040 | 01-01-2023 | 210939 CUST12432  | Manish Mishra     | Office Supplies | Paper       | A4 Paper Ream          | East         | Odisha         | Bhubaneswar | 5    | 0.1      | 410.64   | 1847.88    | 283.38   | Credit Card | Corporate      | Online           | 2023          | Q1   | Jan     |       |
| ORD131145 | 01-01-2023 | 213825 CUST1003   | Nandini Yadav     | Furniture       | Tables      | Study Table            | North        | Punjab         | Ludhiana    | 2    | 0.15     | 6655.64  | 11314.59   | 919.09   | Debit Card  | Small Business | Marketplace      | 2023          | Q1   | Jan     |       |
| ORD130303 | 01-01-2023 | 221646 CUST10841  | Ayush Kumar       | Fashion         | Women       | Women's Kurti          | South        | Telangana      | Hyderabad   | 6    | 0.1      | 1276.22  | 6891.59    | 1153.29  | Net Banking | Consumer       | Online           | 2023          | Q1   | Jan     |       |
| ORD118442 | 01-01-2023 | 223537 CUST116037 | Meera Verma       | Electronics     | Accessories | Bluetooth Speaker      | West         | Goa            | Panaji      | 3    | 0.2      | 3666.33  | 8799.19    | 147.85   | Cash        | Small Business | Marketplace      | 2023          | Q1   | Jan     |       |
| ORD147236 | 01-01-2023 | 224920 CUST15321  | Aditya Srivastava | Electronics     | Laptops     | Lenovo IdeaPad Slim    | South        | Tamil Nadu     | Chennai     | 3    | 0.1      | 48160.12 | 130023.3   | 8751.18  | Credit Card | Consumer       | Retail Store     | 2023          | Q1   | Jan     |       |
| ORD130721 | 01-01-2023 | 233531 CUST11677  | Nandini Shah      | Furniture       | Chairs      | Ergonomic Office Chair | Central      | Madhya Pradesh | Bhopal      | 10   | 0.25     | 8282.93  | 62121.98   | 2357.08  | Debit Card  | Consumer       | Retail Store     | 2023          | Q1   | Jan     |       |
| ORD138065 | 01-01-2023 | 234051 CUST14106  | Ishita Srivastava | Fashion         | Footwear    | Running Shoes          | Central      | Chhattisgarh   | Raipur      | 4    | 0.15     | 3027.76  | 10294.38   | 1513.27  | UPI         | Consumer       | Online           | 2023          | Q1   | Jan     |       |
| ORD118727 | 02-01-2023 | 000634 CUST14077  | Arijun Das        | Home Appliance  | Kitchen     | Mixer Grinder          | North        | Uttar Pradesh  | Kanpur      | 7    | 0        | 4139.24  | 28974.68   | 4048.07  | UPI         | Consumer       | Retail Store     | 2023          | Q1   | Jan     |       |
| ORD120401 | 02-01-2023 | 002628 CUST16984  | Kavya Singh       | Electronics     | Laptops     | HP Pavilion 15         | North        | Uttar Pradesh  | Kanpur      | 4    | 0.1      | 65542.57 | 235953.3   | 15030.22 | Credit Card | Consumer       | Retail Store     | 2023          | Q1   | Jan     |       |
| ORD136846 | 02-01-2023 | 003534 CUST12384  | Nandini Malhotra  | Fashion         | Footwear    | Running Shoes          | South        | Karnataka      | Bengaluru   | 3    | 0.1      | 3448.49  | 5501.31    | 1054.48  | Net Banking | Consumer       | Online           | 2023          | Q1   | Jan     |       |
| ORD100976 | 02-01-2023 | 010901 CUST13849  | Simran Verma      | Electronics     | Laptops     | Lenovo IdeaPad Slim    | North        | Haryana        | Gurgaon     | 2    | 0        | 5353.63  | 107043.3   | 16441.84 | Debit Card  | Consumer       | Retail Store     | 2023          | Q1   | Jan     |       |
| ORD113866 | 02-01-2023 | 011131 CUST11263  | Vivekan Bansal    | Electronics     | Laptops     | Lenovo IdeaPad Slim    | North        | Haryana        | Gurgaon     | 3    | 0.1      | 56580.32 | 152766.9   | 12710.20 | Cash        | Corporate      | Online           | 2023          | Q1   | Jan     |       |

Excel Clean Data

## Appendix B — SQL Database

|    | Order_ID<br>text | Order_Date<br>date | Customer_ID<br>text | Customer_Name<br>text | Category<br>text | Sub_Category<br>text | Product_Name<br>text   | Region<br>text | State<br>text   | City<br>text | Quantity<br>bigint | Discount<br>double precision | Unit_Pri<br>double |
|----|------------------|--------------------|---------------------|-----------------------|------------------|----------------------|------------------------|----------------|-----------------|--------------|--------------------|------------------------------|--------------------|
| 1  | ORD1058...       | 2023-01-01         | CUST11884           | Arjun Bansal          | Electronics      | Accessories          | Power Bank 20000m...   | East           | Bihar           | Patna        | 1                  | 0.1                          |                    |
| 2  | ORD1304...       | 2023-01-01         | CUST17134           | Sneha Agarwal         | Electronics      | Accessories          | Power Bank 20000m...   | South          | Telangana       | Hyderabad    | 4                  | 0.2                          |                    |
| 3  | ORD1444...       | 2023-01-01         | CUST13701           | Nandini Verma         | Furniture        | Storage              | Bookshelf              | North          | Uttarakhand     | Dehradun     | 8                  | 0.05                         |                    |
| 4  | ORD1429...       | 2023-01-01         | CUST15691           | Manish Pandey         | Furniture        | Storage              | Filing Cabinet         | East           | Odisha          | Bhubanesw... | 2                  | 0                            |                    |
| 5  | ORD1268...       | 2023-01-01         | CUST14335           | Neha Malhotra         | Electronics      | Mobiles              | iPhone 15              | South          | Karnataka       | Bengaluru    | 1                  | 0.05                         |                    |
| 6  | ORD1223...       | 2023-01-01         | CUST15049           | Sneha Das             | Electronics      | Laptops              | HP Pavilion 15         | West           | Goa             | Panaji       | 2                  | 0                            |                    |
| 7  | ORD1300...       | 2023-01-01         | CUST16003           | Raj Verma             | Electronics      | Mobiles              | OnePlus Nord CE        | West           | Goa             | Panaji       | 4                  | 0.15                         |                    |
| 8  | ORD1490...       | 2023-01-01         | CUST17928           | Rahul Kumar           | Fashion          | Footwear             | Running Shoes          | North          | Punjab          | Ludhiana     | 3                  | 0                            |                    |
| 9  | ORD1161...       | 2023-01-01         | CUST10899           | Rahul Mehta           | Furniture        | Tables               | Study Table            | East           | Jharkhand       | Ranchi       | 1                  | 0.15                         |                    |
| 10 | ORD1264...       | 2023-01-01         | CUST16374           | Simran Agarwal        | Furniture        | Chairs               | Ergonomic Office Ch... | West           | Gujarat         | Ahmedabad    | 2                  | 0.15                         |                    |
| 11 | ORD1345...       | 2023-01-01         | CUST13140           | Vivaan Bansal         | Office Suppli... | Stationery           | Premium Notebook       | East           | Bihar           | Patna        | 1                  | 0.15                         |                    |
| 12 | ORD1393...       | 2023-01-01         | CUST16915           | Raj Choudhary         | Home Applia...   | Kitchen              | Air Fryer              | West           | Goa             | Panaji       | 4                  | 0.2                          |                    |
| 13 | ORD1145...       | 2023-01-01         | CUST17226           | Simran Sharma         | Fashion          | Footwear             | Running Shoes          | East           | West Bengal     | Kolkata      | 3                  | 0                            |                    |
| 14 | ORD1444...       | 2023-01-01         | CUST12944           | Neha Mishra           | Home Applia...   | Kitchen              | Air Fryer              | West           | Maharashtra     | Mumbai       | 10                 | 0.05                         |                    |
| 15 | ORD1434...       | 2023-01-01         | CUST15409           | Ananya Joshi          | Furniture        | Tables               | Office Work Desk       | South          | Telangana       | Hyderabad    | 1                  | 0                            |                    |
| 16 | ORD1242...       | 2023-01-01         | CUST15975           | Pooja Bansal          | Electronics      | Laptops              | HP Pavilion 15         | East           | Bihar           | Patna        | 1                  | 0.05                         |                    |
| 17 | ORD1052...       | 2023-01-01         | CUST11506           | Rohan Joshi           | Fashion          | Women                | Women's Kurti          | West           | Maharashtra     | Mumbai       | 2                  | 0                            |                    |
| 18 | ORD1431...       | 2023-01-01         | CUST14352           | Priya Mehta           | Home Applia...   | Kitchen              | Air Fryer              | South          | Andhra Prade... | Vijayawada   | 1                  | 0.15                         |                    |
| 19 | ORD1381...       | 2023-01-01         | CUST15812           | Riya Sharma           | Electronics      | Accessories          | Bluetooth Speaker      | South          | Kerala          | Kochi        | 1                  | 0.2                          |                    |
| 20 | ORD1358...       | 2023-01-01         | CUST14895           | Vivaan Sharma         | Office Suppli... | Stationery           | Ball Pen Pack          | West           | Gujarat         | Surat        | 2                  | 0.1                          |                    |
| 21 | ORD1370...       | 2023-01-01         | CUST14483           | Manish Malhotra       | Electronics      | Accessories          | Wireless Earbuds       | North          | Uttar Pradesh   | Kanpur       | 4                  | 0.3                          |                    |
| 22 | ORD1336...       | 2023-01-01         | CUST14149           | Priya Malhotra        | Electronics      | Laptops              | Lenovo IdeaPad Slim    | West           | Gujarat         | Surat        | 3                  | 0.05                         |                    |
| 23 | ORD1030...       | 2023-01-01         | CUST17122           | Neha Kumar            | Office Suppli... | Paper                | A4 Paper Ream          | North          | Uttar Pradesh   | Kanpur       | 2                  | 0.05                         |                    |

### Cleaned Sales Dataset Stored in SQL Database

## Appendix C — Python & NumPy Analysis



### Python-Based Sales Analysis

data.describe(include="object")Python

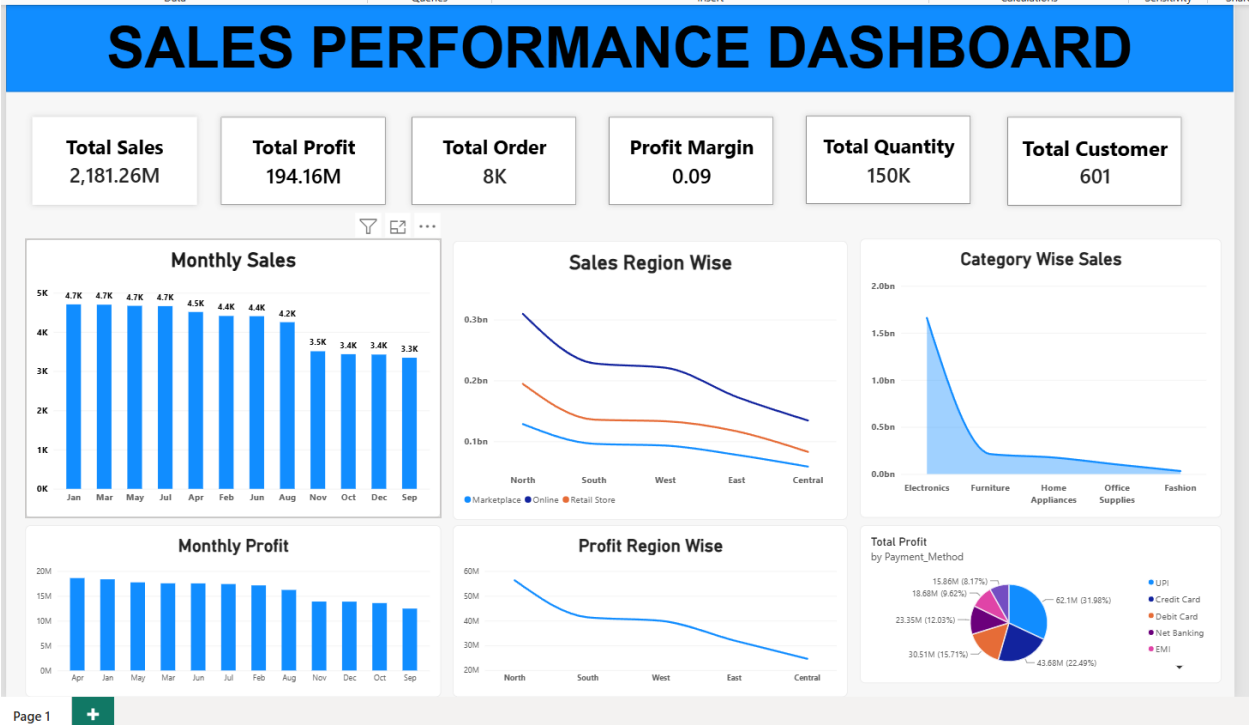
|        | Order_ID  | Customer_ID | Customer_Name | Category    | Sub_Category | Product_Name   | Region | State          | City   | Payment_Method | Customer_Segment | Sales_Channel | Quarter | Month | Year  |
|--------|-----------|-------------|---------------|-------------|--------------|----------------|--------|----------------|--------|----------------|------------------|---------------|---------|-------|-------|
| count  | 50000     | 50000       | 50000         | 50000       | 50000        | 50000          | 50000  | 50000          | 50000  | 50000          | 50000            | 50000         | 50000   | 50000 | 50000 |
| unique | 50000     | 7985        | 600           | 5           | 15           | 25             | 5      | 20             | 24     | 6              | 3                | 3             | 4       | 12    |       |
| top    | ORD116464 | CUST16756   | Shreya Singh  | Electronics | Laptops      | HP Pavilion 15 | North  | Madhya Pradesh | Raipur | UPI            | Consumer         | Online        | Q1      | Jan   | 2     |
| freq   | 1         | 18          | 124           | 17992       | 6079         | 2074           | 14399  | 4177           | 2179   | 16044          | 27657            | 24955         | 13807   | 4703  |       |

data.describe(include="all")Python

|        | Order_ID  | Order_Date                    | Customer_ID | Customer_Name | Category    | Sub_Category | Product_Name   | Region | State          | City   | Quantity     | Discount     | Unit_Price   | Sales        |
|--------|-----------|-------------------------------|-------------|---------------|-------------|--------------|----------------|--------|----------------|--------|--------------|--------------|--------------|--------------|
| count  | 50000     | 50000                         | 50000       | 50000         | 50000       | 50000        | 50000          | 50000  | 50000          | 50000  | 50000.000000 | 50000.000000 | 50000.000000 | 50000.000000 |
| unique | 50000     | NaN                           | 7985        | 600           | 5           | 15           | 25             | 5      | 20             | 24     | NaN          | NaN          | NaN          | NaN          |
| top    | ORD116464 | NaN                           | CUST16756   | Shreya Singh  | Electronics | Laptops      | HP Pavilion 15 | North  | Madhya Pradesh | Raipur | NaN          | NaN          | NaN          | NaN          |
| freq   | 1         | NaN                           | 18          | 124           | 17992       | 6079         | 2074           | 14399  | 4177           | 2179   | NaN          | NaN          | NaN          | NaN          |
| mean   | NaN       | 2024-10-25 18:24:48.073440256 | NaN         | NaN           | NaN         | NaN          | NaN            | NaN    | NaN            | NaN    | 2.993520     | 0.101938     | 16139.241250 | 43625.191153 |
| min    | NaN       | 2023-01-01 00:16:58           | NaN         | NaN           | NaN         | NaN          | NaN            | NaN    | NaN            | NaN    | 1.000000     | 0.000000     | 162.040000   | 120.710000   |
| 25%    | NaN       | 2023-11-26 11:33:51.500000    | NaN         | NaN           | NaN         | NaN          | NaN            | NaN    | NaN            | NaN    | 1.000000     | 0.050000     | 2102.390000  | 4124.057500  |
| 50%    | NaN       | 2024-10-28 21:40:10           | NaN         | NaN           | NaN         | NaN          | NaN            | NaN    | NaN            | NaN    | 2.000000     | 0.100000     | 6955.830000  | 13728.880000 |

Python-Based Sales Analysis

Appendix D — Power BI Dashboard



Sales Performance Power BI Dashboard